first name | last name | email @uwlax.edu

f26.207.quiz08::30

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Find the critical numbers of $f(x) = x^3 - 3x$

To find critical numbers of f
- Where is f' equal to 0?
- Where is f' undefined?

$f'(x) = 3x^2 - 3$ always defined

$3x^2 - 3 = 0$

$3(x^2 - 1) = 0$

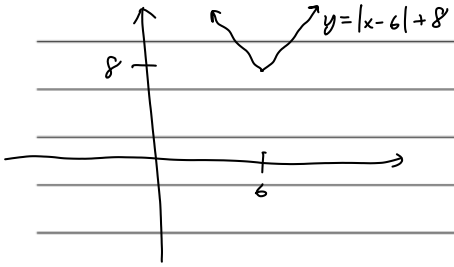
$3(x+1)(x-1) = 0$

$3 = 0$ | $x+1=0$ | $x-1=0$

No sol. | $x = -1$ | $x = 1$

-1 and 1 are the only critical numbers of f.

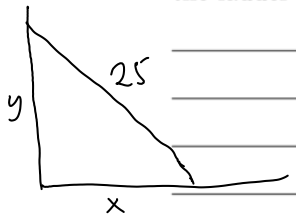
(10pts) 2. Find the critical numbers of $f(x) = |x - 6| + 8$ from a graph.



$f'(6)$ dne

6 is the only critical number of f

(10pts) 3. A 25-foot ladder slides down a wall. The point where the ladder touches the wall moves at the rate of 2 feet per minute. At the moment that the bottom of the ladder is 7 feet from the wall, how fast is the bottom of the ladder moving? Fact: $7^2 + 24^2 = 25^2$.



$\frac{dy}{dt} = -2$

Find $\frac{dx}{dt}$ when $x = 7$

$x^2 + y^2 = 25^2$

$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$

Side: $7^2 + y^2 = 25^2$

$y = 24$

$2(7) \cdot \frac{dx}{dt} + 2(24)(-2) = 0$

$2 \cdot 7 \cdot \frac{dx}{dt} - 2 \cdot 24 \cdot 2 = 0$

$2 \cdot 7 \cdot \frac{dx}{dt} = 2 \cdot 24 \cdot 2$

$\frac{dx}{dt} = \frac{2 \cdot 24 \cdot 2}{2 \cdot 7}$ ft/min